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Original article

Perturbation response scanning of drug-target networks: Drug repurposing for multiple sclerosis

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ABSTRACT

Combined with elastic network model (ENM), the perturbation response scanning (PRS) has emerged as a robust technique for pinpointing allosteric interactions within proteins. Here, we proposed the PRS analysis of drug-target networks (DTNs), which could provide a promising avenue in network medicine. We demonstrated the utility of the method by introducing a deep learning and network perturbation-based framework, for drug repurposing of multiple sclerosis (MS). First, the MS comorbidity network was constructed by performing a random walk with restart algorithm based on shared genes between MS and other diseases as seed nodes. Then, based on topological analysis and functional annotation, the neurotransmission module was identified as the "therapeutic module" of MS. Further, perturbation scores of drugs on the module were calculated by constructing the DTN and introducing the PRS analysis, giving a list of repurposable drugs for MS. Mechanism of action analysis both at pathway and structural levels screened dihydroergocristine as a candidate drug of MS by targeting a serotonin receptor of serotonin 2B receptor (HTR2B). Finally, we established a cuprizone-induced chronic mouse model to evaluate the alteration of HTR2B in mouse brain regions and observed that HTR2B was significantly reduced in the cuprizone-induced mouse cortex. These findings proved that the network perturbation modeling is a promising avenue for drug repurposing of MS. As a useful systematic method, our approach can also be used to discover the new molecular mechanism and provide effective candidate drugs for other complex diseases.

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1. Introduction

The rapid evolution of network medicine has opened a door to understand disease pathogenesis and drug discovery at a broader, networked level [1]. Gaining prominence in drug repurposing, network proximity and network module-based methods allow for the effective identification and comparison of potential repurposed drugs by evaluating the interaction distances between drugs and disease modules, which have demonstrated crucial advantages in devising treatment strategies for complex diseases such as SARS-CoV-2, cancer, and schizophrenia [2–6]. Additionally, the occurrence of diseases is modeled as a network rewiring process, where the drug effects are interpreted as local perturbations of the human interactome network [7]. Predicting perturbation patterns based on the topology of biological networks [8] provide a suitable framework for describing disease phenotypes and recovering potential repurposable drugs. Accordingly, the principles of network modules and network perturbation, combined with deep learning methods, can guide drug repurposing efforts [9,10]. However, new methods that consider the physical and dynamical nature of modeling chemical perturbations are still needed.

Towards drug repositioning requires not only understanding the direct relationships between drugs and disease, but also indirect relationships between diseases, and dynamical interactions between drugs and targets. However, indirect and dynamical cooperations have been notoriously hard to quantify. To this aim, biophysical methods provide complementary tools for modeling network dynamics. Elastic network models (ENMs) have long provided a rigorous framework for describing protein dynamics defined by their contact topology [11,12]. In parallel, the perturbation response scanning (PRS) approach, rooted in linear response theory, allows evaluating network dynamics in response to external forces by calculating the effectiveness and sensitivity of nodes. Combined with ENMs, PRS was initially developed for modeling protein dynamics and identifying communication interactions within proteins [13,14]. It's worth noting that signal propagation does not simply halt at the confines of a protein, rather, it can dynamically traverse the cell landscape [15]. The recent advancement has demonstrated that PRS can be extended beyond protein structures to system-level networks by leveraging the principles of perturbation propagation. PRS provided an exact mathematical framework to model the yeast genetic interaction profile similarity network, by studying high-level organization and cooperativity within the cellular network [16]. When applying PRS to drug-target networks (DTNs) as a fundamental model in drug repurposing, it can assess the perturbations induced by a drug on its target, considering network dynamics [17].

In this work, we developed a deep learning and dynamical network-based drug repurposing approach, namely DTNPerturbome, bridging module identification in network biology and PRS in computational biophysics. Our approach focused on the analysis of DTN of multiple sclerosis (MS), a chronic inflammatory and degenerative disease of the central nervous system (CNS) [18]. Despite substantial research on the mechanisms of MS pathology [19,20] and the discovery of novel therapeutic strategies [21,22], only a few drugs are available to alleviate symptoms, and no definitive cure exists, making MS treatment a challenging medical endeavor. The framework of DTNPerturbome for MS contains three steps: (1) constructing the MS comorbidity network by performing a random walk with restart algorithm (RWR) based on shared genes between MS and other diseases as seed nodes, (2) identifying the MS-related therapeutic module by the topological analysis and functional annotation, and (3) ranking the drugs for MS based on the therapeutic module, by introducing the DeepPurpose [23] and PRS to construct and analyze the DTN. The proposed repurposable drugs and potential targets for MS were screened by mechanism of action analysis, followed by an investigation of the role of potential targets in the cuprizone-induced mouse model of MS.

2. Methods

2.1. Data collection

MS-related genes were collected from five databases including DisGeNET, DISEASES, OMIM, PheGenI, and PGKB, with detailed information listed in Table S1. To identify shared genes of MS and other diseases, we collected the disease genes from all 298 diseases (excluding MS) recorded in the Drug Repurposing Hub database [24]. Specifically, the latest version of the drug information (dated March 24, 2020) was downloaded, which includes details on compound names, clinical stages, mechanisms of action, protein targets, and associated diseases. Entries with indications listed as empty or only as "multiple sclerosis" were excluded, resulting in records for drugs and targets related to 298 diseases. After removing duplicate target genes, a final list of 2161 unique disease genes was obtained, which, along with MS-related genes, were used to calculate the shared genes.

The drug-target interaction (DTI) data was obtained from DrugBank [25] and Therapeutic Target Database [26]. The human protein-protein interaction (PPI) data was gathered from the STRING database [27]. In order to balance PPI confidence and network size, interactions with a combined score $\geq$400 were extracted from the global interactome, which ensures high-quality interactions while excluding those that are less likely to be biologically. The blood-brain barrier permeability (BBBP) of drugs was predicted using admetSAR [28] and retrieved from the DrugBank database.

2.2. Comorbidity network generation

The generation of MS comorbidity network contains three steps. The first step is the construction of the seed PPI network (PPIN), in which nodes are interconnected proteins derived from the intersection of MS and other disease genes after mapping onto the human PPIN. The second step is the generation of MS network by applying the RWR algorithm [29] to the seed PPIN. From the seed proteins as source node s, an iterative walker moves from its present node to a randomly chosen neighboring node. At each step, there's a probability r for the walk to restart at node s. Formally, the random walk with restart is defined as:

$$p^{t+1} = (1-r)Wp^t + rp^0 \quad (1)$$

where W is the column-normalized adjacency matrix of the graph, r represents the restart probability, p^0 is the initial weight vector based on a particular node, and p^t is a vector in which the ith element holds the probability of being at node i at time step t. By implementing the RWR algorithm with a restart probability of 0.8 [30], the probabilities of both seed and non-seed nodes to visit other nodes were calculated. The last step is to obtain "reachable proteins" by computing the "reachability" between any two nodes on the network [31]. To identify nodes that are topologically closer to the seed proteins, the access probability scores of the ranked nodes on the network (excluding seed proteins) were divided into two groups, namely those of accessing the seed protein group (S_{seed}) and the non-seed protein group ($S_{non-seed}$), with the fold change defined as:

$$Fold\ change = \frac{\overline{S}_{seed}}{\overline{S}_{non-seed}} \quad (2)$$

where $\overline{S}_{seed}$ and $\overline{S}_{non-seed}$ represent the average accessibility scores of rank nodes accessing seed proteins and non-seed proteins,

respectively. The ratio indicates whether non-seed nodes are more closely related to seed nodes or non-seed nodes. A higher fold change suggests that non-seed nodes are more closely associated with seed nodes, implying potential functional or biological relevance. Additionally, the Wilcoxon rank-sum test on the accessibility scores of the two groups was performed, which enabled us to ascertain whether specific nodes displayed significant reachability to seed proteins compared to their connectivity with other proteins in the network. Proteins with the *P*-value less than 0.05, and the log2 (fold change) greater than 2 are identified as "reachable proteins". The MS comorbidity network was generated by expanding the original seed PPIN until it encompassed over 90% of the "reachable proteins", after which module division and functional annotation methods were applied to the network (See Supplementary Methods for more details).

2.3. Network topology analysis

In addition to degree for nodes in a network, two network measures including within-module degree (Z) and participation coefficient (P) were calculated to assess the role of each node within its module and its interaction with other modules [32]. Z measures the number of intramodule links of a node in relation to the average number of links of nodes within the module, which is defined as:

$$Z_i = \frac{k_i - \overline{k}_{s_i}}{\sigma_{k_{s_i}}} \tag{3}$$

where k_i is the number of links of node i, and $\overline{k}_{s_i}$ and $\sigma_{k_{s_i}}$ are the average and standard deviation of the degree of nodes in module s_i respectively. P quantifies the distribution of links of a node across different modules, which can be computed by:

$$P_i = 1 - \sum_{s=1}^{N_M} \left(\frac{k_{is}}{k_i}\right)^2 \tag{4}$$

where k_{is} is the number of links of node i in module s, and N_M is the total number of modules. All nodes are divided into peripherals, connectors, module hubs, and network hubs based on predetermined thresholds [33]. The peripheral nodes have both a low $Z_i \leq 2.5$ and a low $P_i \leq 0.62$, with a few links inside its own module and rarely to other modules. The connectors, having a low $Z_i \leq 2.5$ and a high $P_i > 0.62$, glue modules together and are thus important to network coherence. A node with a high $Z_i > 2.5$ and a low $P_i \leq 0.62$ is defined as module hub that is important to the coherence of its own module. A network hub has both a high $Z_i > 2.5$ and a high $P_i > 0.62$, being important to the coherence of both the network and its own module.

2.4. PRS of drug-target network

To predict and compare the effects of different drugs on a network, we employed a PRS-based approach based on DTNs. This quantitative method integrates the binding affinity of drug-target pairs and the propagation of their impact through protein interactions within the network. Overall, it consists of four steps (Fig. 1). First, an initial force is assigned to a specific protein, with the force magnitude determined by the predicted binding affinity between the drug and its target protein on the DTN. Then, a unit force is applied and transmitted through the network from one protein to another, generating an average impact. This impact is estimated using the Gaussian Network Model (GNM), where proteins are represented as nodes, interactions as springs, and network junctions exhibit Gaussian fluctuations. Next, the propagated force is incorporated to evaluate the drug's effect on the entire network through its interaction with the target protein. Finally, for drugs targeting multiple proteins, individual impacts are aggregated to determine the overall effect on the network module. Detailed procedures for each step are provided below.

Step 1. Constructing a heterogeneous network of the weighted DTN. The topology of the DTN was defined by adding the collected drugs and the corresponding drug-target interactions (DTIs) as new nodes and edges to the PPIN. The edges of DTNs were defined by

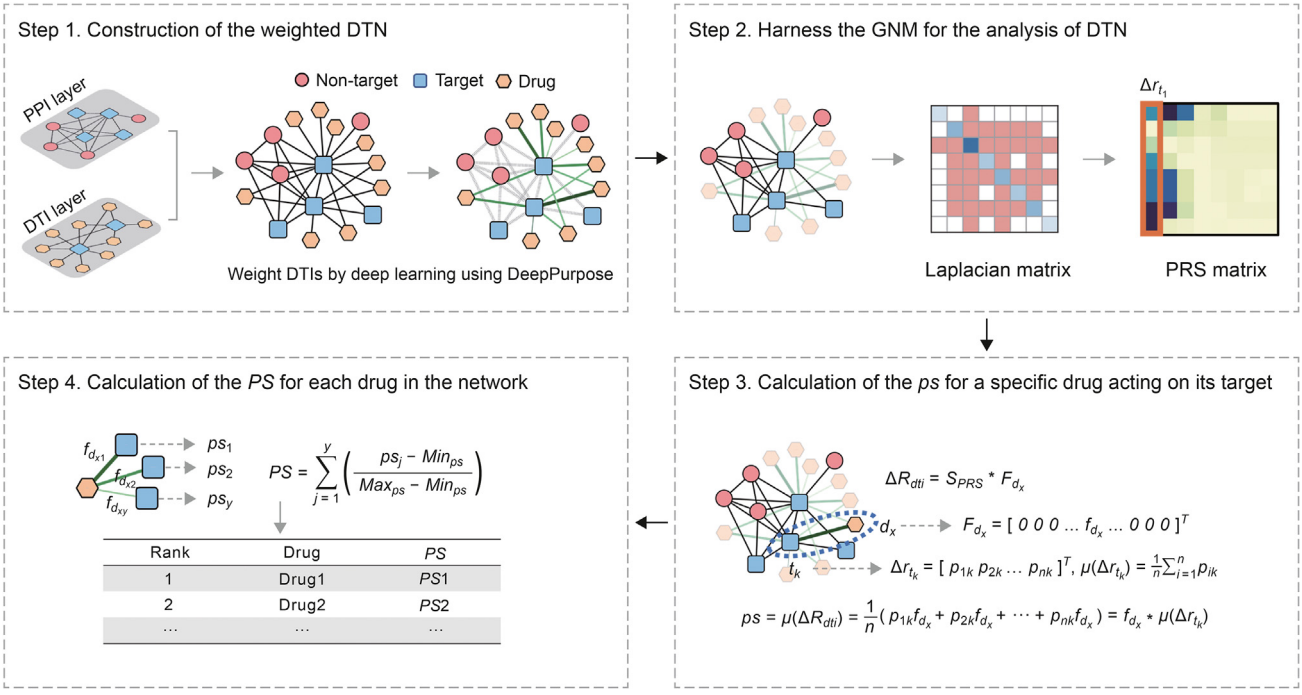


Fig. 1. The overview of the perturbation response scanning (PRS)-based drug repurposing method. DTN: drug-target network; DTI: drug-target interaction; GNM: Gaussian Network Model; *ps*: perturbation score; *PS*: the cumulative score of the normalized *ps* of drugs targeting multiple proteins.

binding affinities based on the deep-learning method of DeepPurpose [23], which employs a DTI model as an encoder-decoder framework, with eight encoders for drugs, namely Morgan, PubChem, Daylight, RDKit 2D, convolutional neural network (CNN), convolutional recurrent neural network (CNN + RNN), Transformer encoders, and message-passing neural network (MPNN), and seven protein target encoders including amino acid composition (AAC), PseACC, conjoint triad, quasi sequence, CNN, CNN + RNN, and Transformer. Taking the simplified molecular-input line-entry system (SMILES) format of drugs and protein amino acid sequences as input, DeepPurpose outputs scores on behalf of the binding affinity between drugs and targets. Taking the sequences and SMILES formats as inputs, we utilized the pre-trained model MPNN_CNN_BindingDB_IC50 provided by DeepPurpose to predict the binding affinity between each DTI.

Step 2. We then harnessed the GNM [11], originally designed to elucidate intrinsic properties within protein structures, to analyze the PPI layer in the DTN. This involved envisioning the protein interaction component as a network structure meticulously crafted from interconnected elastic beads and springs. The global network connectivity is depicted through a Laplacian matrix, where diagonal elements represent node degrees, and non-zero off-diagonal elements signify pairwise node connections. Within this context, the PRS calculation emerges as a method enabling us to assess how nodes within the system respond to external disturbances [13,14].

PRS was initially proposed within the framework of GNMs for structural calculations. It involves constructing a vector by applying a force (f_i) at each node, with all entries set to zero except for the entry corresponding to the perturbed node. For a selected node i, the external force vector is constructed as follows:

$$(f_i)^T = \left\{0\ 0\ 0 \ldots f_i^x\ f_i^y\ f_i^z \ldots 0\ 0\ 0\right\}_{1\times 3n} \tag{5}$$

Under equilibrium conditions, the positional displacements of all n nodes in the system are determined by the force balance. These $3n$-dimensional displacements (Δr_i) can be assessed using the known pseudo-inverse of the $3n \times 3n$ Hessian matrix H:

$$\Delta \acute{r}_i = H^{-1} f_i. \tag{6}$$

In our method, applying models originally designed for atomic networks to molecular networks, under isotropic conditions represented by the GNM, treats the protein interaction network as a structure of interconnected springs. The overall connectivity of this network is described by a Laplacian matrix. Here, both external force and displacement are simplified to n-dimensional vectors, with $\Delta \acute{r}_i$ and f_i representing the displacement magnitudes and forces corresponding to n nodes, respectively. Then, H is represented by the $n \times n$ symmetric Laplacian matrix, Γ.

$$\Delta \tilde{r}_i = \left(\frac{1}{\gamma}\right) \Gamma^{-1} \tilde{f}_i \tag{7}$$

where γ represents the uniform spring constant. Subsequently, the PRS matrix S_{PRS} in the GNM is determined by executing the operation for all nodes.

The results are then normalized to eliminate the intrinsic biases, specifically the node-specific variance σ_i:

$$S_{PRS} = [\Delta \tilde{r}_1 \Delta \tilde{r}_2 \ldots \Delta \tilde{r}_n] = \left(\frac{1}{\gamma}\right) \Gamma^{-1} [diag(\sigma_1\ \sigma_2 \ldots\ \sigma_n)]^{-1} [\tilde{f}_1\ \tilde{f}_2 \ldots \tilde{f}_n] \tag{8}$$

where the diagonal matrix of node-specific variance σ_i, denoted as $[diag(\sigma_1\ \sigma_2 \ldots\ \sigma_n)]^{-1}$, serves as the normalization factor. The element p_{ij} in the S_{PRS} matrix signifies the response of node j to the perturbation at node i. The normalization accounts for adaptability defined by the node's topological structure and disrupts the symmetry of the covariance matrix. Under this assumption, the force applied to each node is consistent. Consequently, the force matrix computed for the PRS matrix is represented as the identity matrix. ProDy is used to compute the PRS matrix [34].

Step 3. To calculate the perturbation score for each drug when acting on its target protein, we represented the drug's affinity for target k as a force vector, $F_{d_x} = [0\ 0\ 0 \ldots f_{d_x} \ldots 0\ 0\ 0]^T$, where all elements are zero except for the kth element which is the specific affinity of the drug for its target protein. This mirrors the method used in protein structures, where a perturbation force is applied at a specific node, allowing us to translate the structural perturbation concept to the network context. Subsequently, we computed the perturbation effect of the drug binding to its target protein by performing a matrix multiplication of the force vector and the PRS matrix S_{PRS}. The result of this operation determines the perturbative effect emerging from a drug binding to its specific target protein on other nodes in the network is denoted as:

$$\Delta R_{dti} = S_{PRS} * F_{d_x} \tag{9}$$

where ΔR_{dti} represents the network-wide displacement vector perturbation caused by the DTI. To calculate the perturbation score (ps) for the drug, we compute the mean perturbation across all nodes in the network for a comprehensive evaluation of the drug's impact. This can be expressed as:

$$ps = \mu(\Delta R_{dti}) \tag{10}$$

This mean perturbation can also be represented as the sum of the products of the perturbation at each node (p_{ik}) and f_{d_x}, divided by the total number of nodes and then the equation is further simplified by taking common from the sum:

$$\begin{aligned} ps &= \frac{1}{n}\left(p_{1k} f_{d_x} + p_{2k} f_{d_x} + \ldots + p_{nk} f_{d_x}\right) = f_{d_x} * \left(\frac{1}{n}\sum_{i=1}^{n} p_{ik}\right) \\ &= f_{d_x} * \mu(\Delta r_{t_k}) \end{aligned} \tag{11}$$

where $\Delta r_{t_k} = [p_{1k}\ p_{2k} \ldots\ p_{nk}]^T$ is the kth displacement vector of PRS matrix S_{PRS} and f_{d_x} is the affinity for a given DTI. This ps score can holistically capture the effect of DTI on the entire DTN.

Step 4. For drugs that interact with multiple target proteins, we aggregate the ps for each DTI to compute the final score for the drug. This can be done by summing the individual normalized ps scores for all DTIs associated with the drug. The PS for a drug acting on multiple target proteins is calculated as:

$$PS = \sum_{j=1}^{y} \frac{ps_j - Min_{ps}}{Max_{ps} - Min_{ps}} \tag{12}$$

where ps_j stands for the jth individual ps score for the drug, Max_{ps} and Min_{ps} are the maximum and minimum ps scores derived from the entire network of DTIs, respectively, and y is the total number of target proteins for the drug. Thus, we can obtain a comprehensive measure of the drug's impact on the network when interacting with multiple target proteins, in part it may also reflect a higher propensity of more highly connected proteins to yield a response upon perturbation, *i.e.*, a therapeutic effect.

2.5. Network proximity calculation

To investigate the proximity between the disease module S (MS seed genes, s) and the drug module T (drug targets t in M2 of a drug) in the enlarged MS PPIN, a network-based proximity measure from SAveRUNNER was implemented [35], defined as:

$$p(T,S) = \frac{1}{|T|}\sum_{t\in T} \min_{s\in S} d(t,s) \quad (13)$$

where p represents the average shortest path length between drug targets t in the drug module T and the nearest disease genes s in the disease module S. The observe network proximity was z-score normalized by applying a degree-preserving randomization procedure. Specifically, we randomly selected protein modules with the same number of nodes and the same degree distribution as the actual target protein module in the target module (M2) of each drug in the enlarged MS PPIN, and then the shortest path distances between these random modules and the disease module were calculated, generating a reference distance distribution. This procedure is repeated 1,000 times and the z statistics is computed by using the mean and the standard deviation of the reference distance distribution.

2.6. Drug and target analysis

Mechanisms of action of candidate drugs based on pathway and structural levels, and experimental validation of the potential target had been carried out. Detailed information is provided in the Supplementary Methods. All animals are taken care according to the National Institutes of Health Laboratory Animal Care and Use Guidelines. Animal experimental protocols are approved by the Animal Ethics Committee of Soochow University (Approval number: SUDA20230824A06).

3. Results

3.1. Generation of MS comorbidity network based on network propagation

By pooling five databases, a total of 845 MS-related disease genes were obtained (Fig. 2A), and then 100 shared genes were further identified by the intersection analysis with 2161 genes related to other diseases (excluding entries for standalone MS or those with missing data) retrieved from the Drug Repurposing Hub (Fig. 2B and Table S2). Through mapping onto the human PPIN, the shared genes generate the seed PPIN of 80 interconnected nodes (Fig. 2C), showing the scale-free property (Fig. 2D). We then implemented RWR propagation method to uncover the "reachable proteins" in the seed PPIN. Based on the reachability to the seed nodes, the ranked proteins produce the MS comorbidity network, which includes 3,680 protein nodes and covers approximately 90% of reachable proteins (Fig. 2E), displaying higher degree centrality

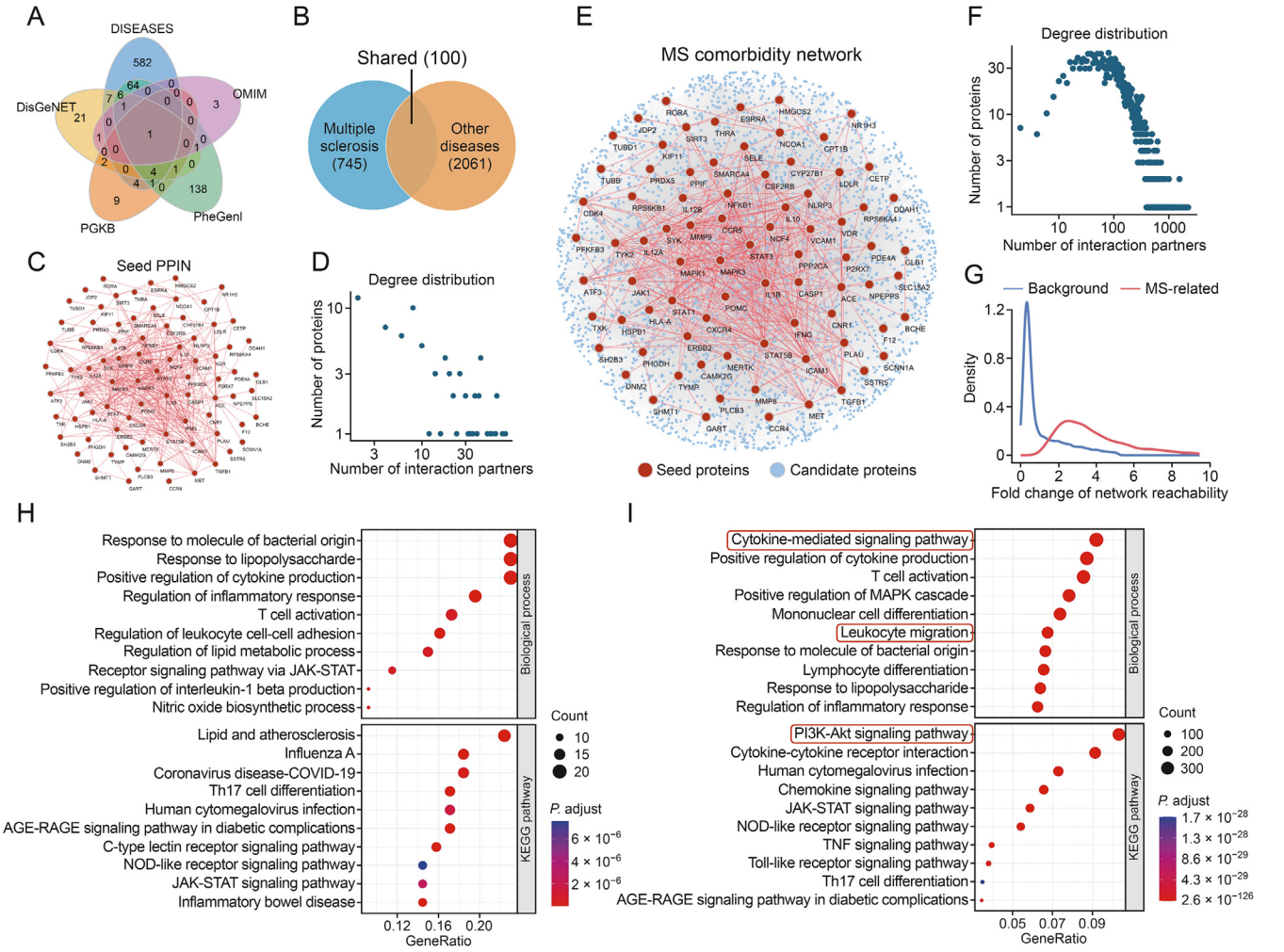


Fig. 2. Generation of the multiple sclerosis (MS) comorbidity network. (A) Venn diagram illustrating the collection of MS disease genes. (B) Venn diagram showing the shared gene count between MS and other diseases. (C) The seed protein-protein interaction network (PPIN) with 80 share genes as nodes, and 406 interactions as edges. (D) Degree distribution of the seed PPIN. (E) The MS comorbidity network obtained by network propagation, with seed proteins highlighted in red and enlarged. (F) Degree distribution of the MS comorbidity network. (G) The reachability of candidate proteins in the MS comorbidity network is significantly higher relative to all other proteins in the human PPIN. (H, I) The top 10 significant entries from the Gene Ontology (GO) biological processes and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment results of the seed network (H) and the MS comorbidity network (I).

(Fig. 2F). By plotting probability density curves (Fig. 2G), new protein nodes in the comorbidity network, compared to the general protein group background, are more likely to interact with the 80 seed proteins in terms of network topology. Additionally, as shown in Fig. S1, the comorbidity network reached a certain scale, with deviations in target gene density occurring only within a limited range compared to the outcomes of random samplings, thus validating that the determined network scale is both reasonable and credible.

Furthermore, we conducted Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway and Gene Ontology (GO) enrichment analyses for both the MS seed and comorbidity networks (Figs. 2H and I) (See Supplementary Methods). Enrichment analysis of both networks revealed a significant association with biological processes and pathways implicated in MS, such as "inflammatory response", "T cell activation", and "regulation of cytokine production". These processes are instrumental to MS, being associated with immune system attack on nerve fibers (causing inflammation),

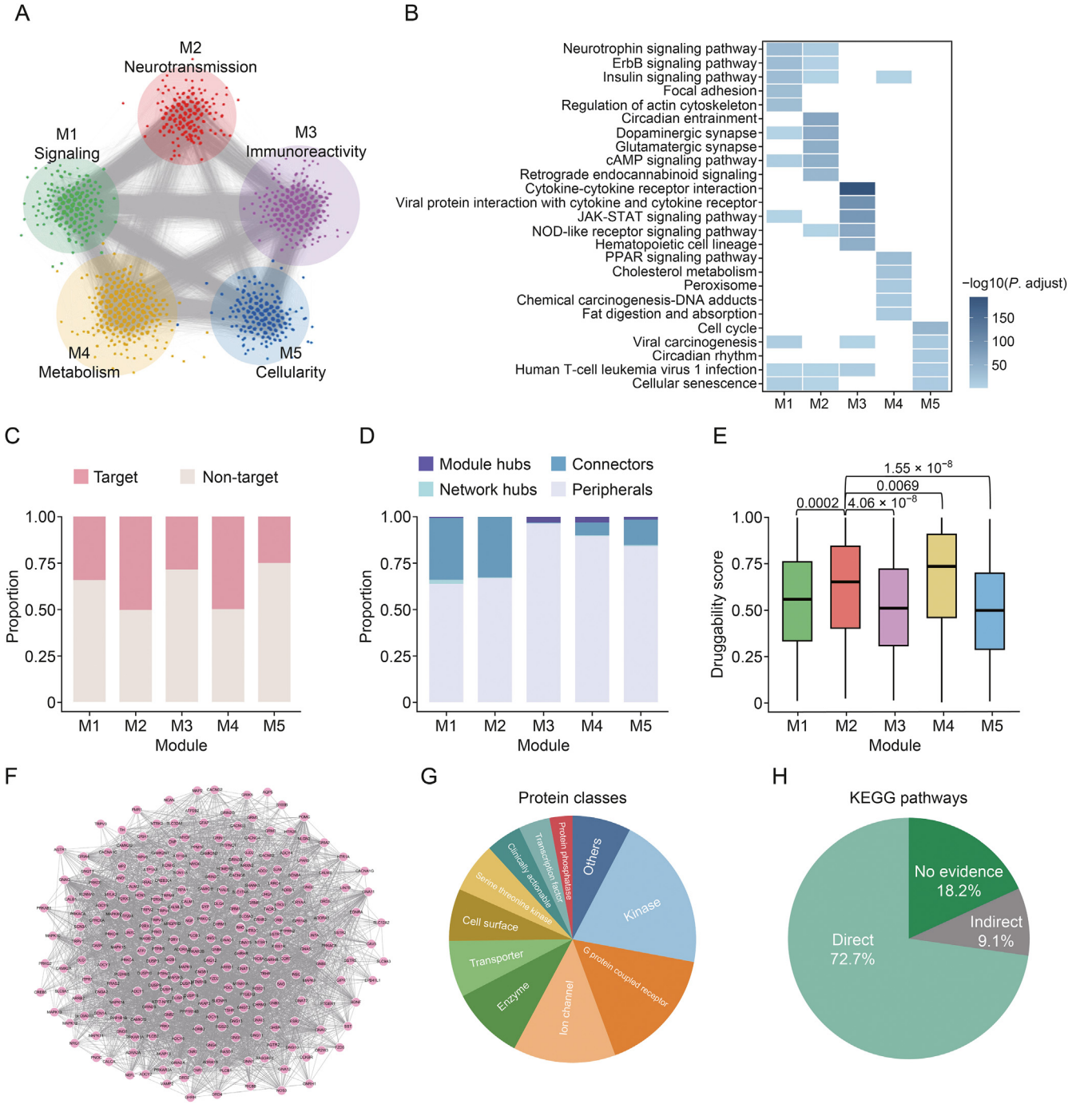


Fig. 3. Identification of M2 as the "therapeutic module" in the multiple sclerosis (MS) comorbidity network. (A) Five functional modules with proteins in each module represented by different colors, whose detailed biological function annotations were found at Text S1. (B) The top five significantly enriched pathways for each module. Each row corresponds to a distinct pathway, while each column represents a unique module. (C) Bar graph showing the enrichment of drug targets in each module. (D) Bar graph depicting the proportions of the four node categories in each module. (E) The distribution of druggability scores for nodes in each module. The significance of druggability scores between M2 and other modules was annotated. (F) The architecture of M2. (G) The protein class ontology results of M2. (H) The relationship of Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways with MS pathogenic mechanisms. See Table S3 for more information.

wrongful activation of immune cells against the body's cells, and imbalance in immune response regulation, respectively [36]. For the MS comorbidity network (Fig. 2I), it uncovered additional MS-associated biological processes, such as "leukocyte migration" and "cytokine-mediated signaling pathway", which represent the migration of immune cells towards inflammation sites and dysregulation of inflammation and immune responses [37]. It also detected additional KEGG pathways like the "PI3K-Akt signaling

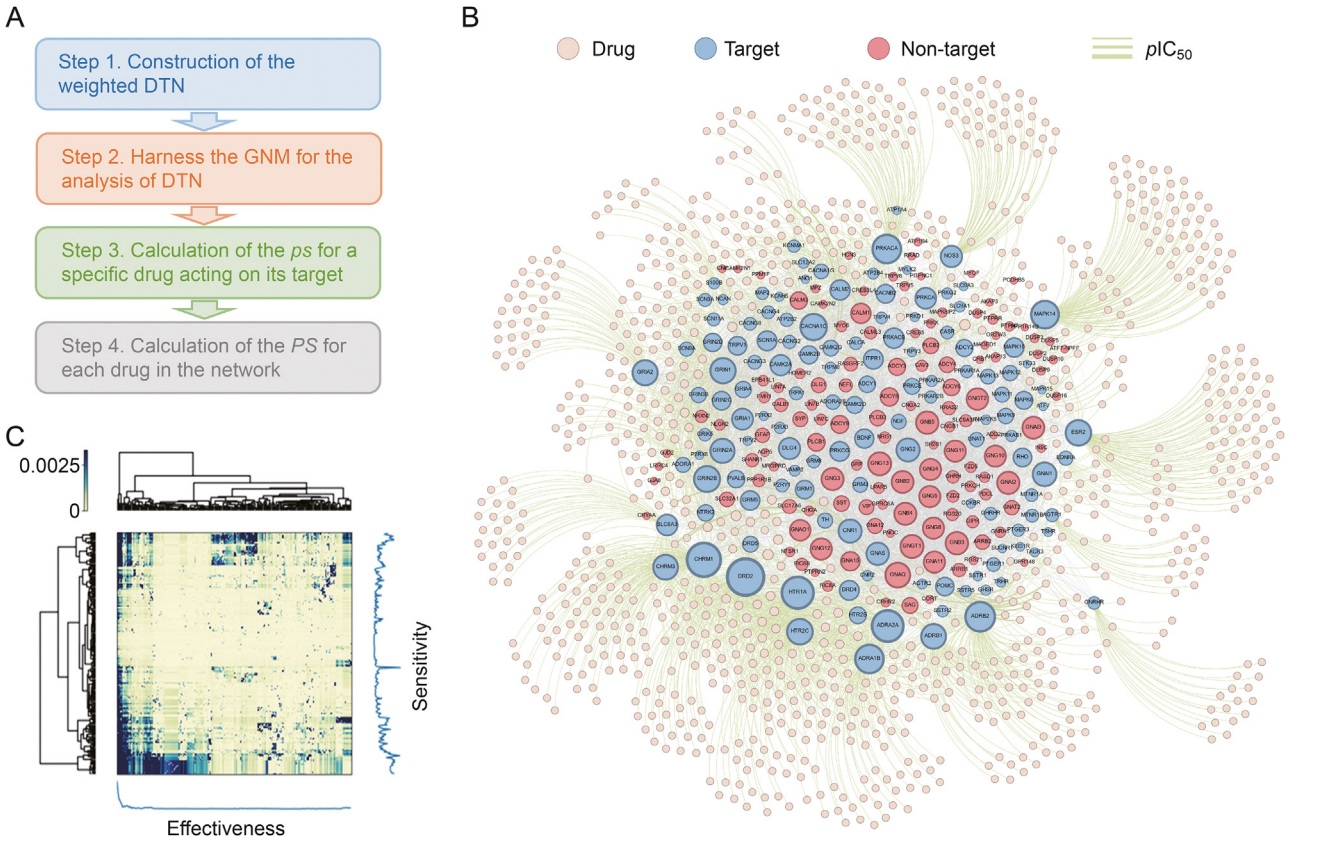


Fig. 4. Drug repurposing through the perturbation response scanning (PRS)-based approach. (A) The pipeline of drug repurposing based on PRS. (B) The weighted drug-target network (DTN) constructed from M2. (C) Visualization heatmap of the calculated PRS matrix for illustrating the propagation of perturbation effects through the identified therapeutic module M2 when specific drugs interact with their target proteins. Row averages and column averages of the PRS matrix indicate the sensitivity (right vertical axis) and effectiveness (bottom horizontal axis) of each node.

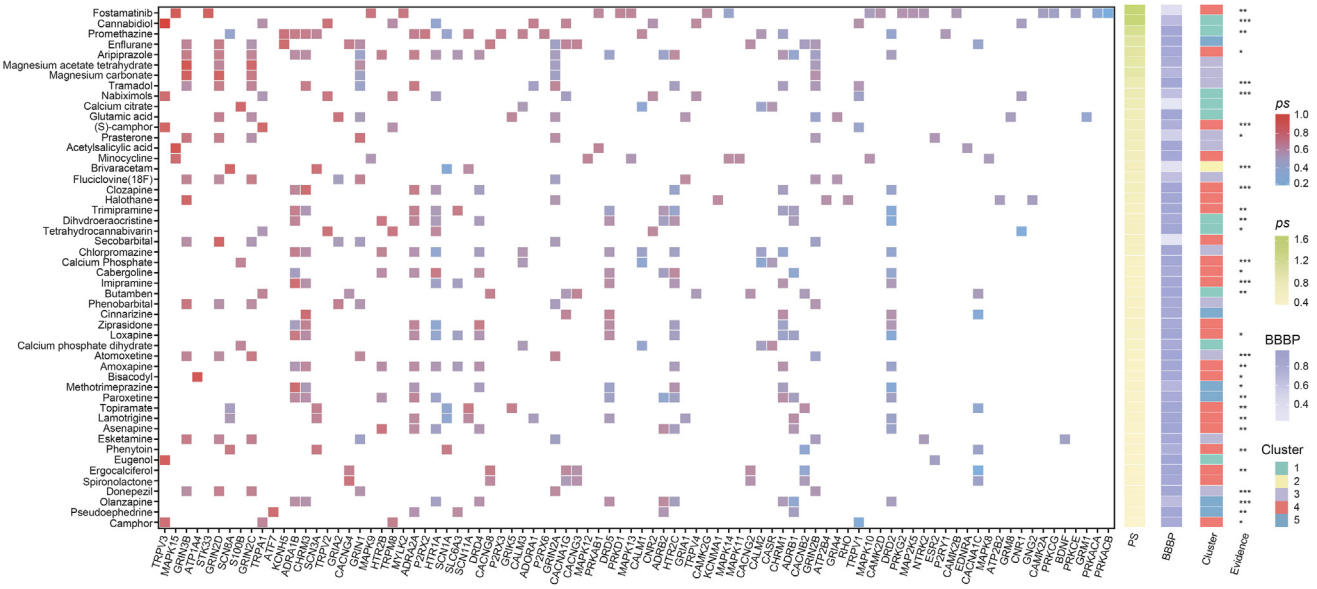


Fig. 5. The heatmap of the top 50 ranked drugs based on their perturbation score (*ps*) scores. Each row corresponds to a drug, and the color gradients along the row represent the drug's *ps* pertaining to each target protein, and the accumulated score of these *ps* along a row yields the cumulative score *PS* for that drug. BBBP: blood-brain barrier permeability.

pathway", indicating the involvement in cell survival and proliferation that are crucial to MS progression [38].

3.2. Identification of the potential therapeutic module

To uncover the molecular mechanism of MS mechanistically, the Node and Edge Prioritization-based Community Analysis (ne-PCA) algorithm [30] was utilized to partition the MS comorbidity network into five distinct functional modules, labeled as M1 through M5 (Fig. 3A) (See Supplementary Methods). Detailed functional annotations of each module by KEGG pathway enrichment analyses were subsequently performed (Fig. 3B).

Then, three aspects of the five modules were further characterized, including 1) the proportion of known drug target proteins, 2) the number of connector nodes by assessing the core-periphery structure within the MS comorbidity network, and 3) the druggability scores of each module, to identify the key therapeutic module. In Fig. 3C, M2 and M4 have higher proportions (almost 50%) of drug targets than other modules. Next, all nodes were classified into four types based on within-module degree and participation coefficient (Fig. S2). Amongst, the connector nodes, which act as "bridges" for communication between different modules, represent potential targets for disease intervention [39]. In Fig. 3D, M1 and M2 have more connector nodes, with a proportion of 33.4% and 32,6%, respectively. Furthermore, the druggability scores of five modules indicate that M2 displays the second highest druggability score of 0.625 (Fig. 3E).

Additionally, M2 consists of 258 nodes and 4,296 edges (Fig. 3F). The protein class ontology analysis of M2 (Fig. 3G) demonstrates a

Table 1
Top 50 repurposed drugs for multiple sclerosis (MS) through perturbation response scanning (PRS)-based approach.

Drugs	*PS*	Original indication	MoA_Cluster	Evidences	BBBP
Fostamatinib	1.691	Chronic immune thrombocytopenia	Cluster 4	**	0.285
Cannabidiol	1.658	Multiple sclerosis, seizures, severe pain, moderate pain	Cluster 1	***	0.712
Promethazine	1.281	Allergic conditions, nausea, vomiting, motion sickness	Cluster 1	**	0.984
Enflurane	1.043	Analgesia, general anesthesia	Cluster 5		0.980
Aripiprazole	0.983	Antipsychotic medication	Cluster 4	*	0.955
Magnesium acetate tetrahydrate	0.927	Laxatives, electrolyte supplementation	Cluster 3		0.879
Magnesium carbonate	0.915	Calculus bladder, dyspepsia, heartburn, kidney stones, upset stomach	Cluster 3		0.781
Tramadol	0.725	Acute, moderate, severe pain	Cluster 3	***	0.975
Nabiximols	0.704	Relieve neuropathic pain in multiple sclerosis	Cluster 1	***	0.672
Calcium citrate	0.676	Over the counter calcium supplement	Cluster 1		0.208
(S)-camphor	0.638	Joint pain, myalgia	Cluster 1		0.989
Acetylsalicylic acid	0.616	Pain, fever, and inflammation	Cluster 4	***	0.850
Glutamic acid	0.607	Amino acid supplementation	Cluster 3	*	0.465
Prasterone	0.596	Moderate dyspareunia, severe dyspareunia	Cluster 3		0.963
Brivaracetam	0.575	Partial-onset seizures	Cluster 4		0.954
Minocycline	0.570	Inflammatory lesions of acne vulgaris	Cluster 2	***	0.252
Fluciclovine (18F)	0.545	Image tumors, especially in the prostate	Cluster 3		0.661
Clozapine	0.536	Antipsychotic medication	Cluster 4	***	0.971
Halothane	0.533	General inhalation anesthetic	Cluster 4		0.974
Trimipramine	0.513	Depression accompanied by anxiety, agitation or sleep disturbance	Cluster 4	**	0.976
Tetrahydrocannabivarin	0.513	One of four cannabinoids, no approved indications	Cluster 1	**	0.931
Calcium Phosphate	0.510	Over the counter calcium and phosphate supplement	Cluster 1	*	0.974
Dihydroergocristine	0.506	Delay progressive mental decline in conditions like Alzheimer's disease	Cluster 4		0.232
Secobarbital	0.502	Insomnia	Cluster 3		0.972
Imipramine	0.479	Attention deficit hyperactivity disorder (ADHD), depression, enuresis	Cluster 4	***	0.982
Cabergoline	0.474	Hyperprolactinemia, idiopathic hyperprolactinemic disorder	Cluster 4	*	0.925
Cinnarizine	0.474	Vertigo, nausea, motion sickness, peripheral arterial disease (PAD)	Cluster 4	***	0.969
Chlorpromazine	0.471	Antipsychotic medication	Cluster 1	**	0.985
Phenobarbital	0.470	All types of seizures except absence seizures	Cluster 3		0.937
Butamben	0.468	Anesthesia of mucus membranes	Cluster 5		0.950
Bisacodyl	0.465	Bowel movement irregularity, constipation	Cluster 4		0.931
Ziprasidone	0.453	Antipsychotic medication	Cluster 4	*	0.977
Calcium phosphate dihydrate	0.450	Over the counter calcium and phosphate supplement	Cluster 1		0.926
Atomoxetine	0.441	ADHD	Cluster 3	***	0.985
Loxapine	0.435	Acute agitation, schizophrenia	Cluster 4	**	0.978
Eugenol	0.428	Contusions, insect bites, myalgia	Cluster 4	*	0.952
Topiramate	0.423	Anticonvulsant	Cluster 5	*	0.782
Phenytoin	0.423	Seizures	Cluster 5	**	0.914
Methotrimeprazine	0.421	Antipsychotic medication	Cluster 4	**	0.976
Amoxapine	0.420	Agitation, anxiety, depression	Cluster 4	**	0.977
Asenapine	0.413	Bipolar 1 disorder, schizophrenia	Cluster 4	**	0.987
Esketamine	0.410	Major depressive disorder (MDD), pain	Cluster 3		0.953
Lamotrigine	0.407	Bipolar 1 disorder, seizures	Cluster 4	**	0.871
Camphor	0.405	Relieve pain, topical antiseptic	Cluster 1		0.990
Paroxetine	0.403	Antipsychotic medication	Cluster 4	**	0.981
Pseudoephedrine	0.400	A sympathomimetic amine used for its decongestant activity	Cluster 4		0.958
Donepezil	0.400	Alzheimer's disease, dementia	Cluster 3	***	0.978
Ergocalciferol	0.400	Deficiency, vitamin A, deficiency, vitamin D, hypoparathyroidism	Cluster 5	***	0.698
Spironolactone	0.400	Edema, hypertension, primary aldosteronism	Cluster 5	**	0.975
Olanzapine	0.378	Acute agitation, Bipolar 1 disorder	Cluster 4	*	0.921

Notes: ***(strong evidence): The corresponding drug research has clearly demonstrated the therapeutic effect of the drug, either through empirical verification in animal models or through significant improvement in disease symptoms in human clinical trials compared to a control group. **(moderate evidence): The corresponding drug research is based on some preliminary evidences, such as theoretical research or bioinformatics analysis, or is only effective in some people in the clinical experiment. *(weak evidence): The corresponding drug research is based on a single case, or there is only evidence showing that the drug has some association with the disease. See Text S2 for the detailed information.
PS: the cumulative score of the normalized *ps* of drugs targeting multiple proteins; BBBP: blood-brain barrier permeability; MoA: mechanism of action.

predominant enrichment of kinases, G protein-coupled receptors (GPCRs), and ion channels, belonging to major protein families as drug targets [40]. For example, GPCRs are the most extensively investigated drug targets, with a notable focus on various CNS disorders [41]. In comparison with other modules, M2 shows stronger biological relevance to MS. Among the enriched pathways of M2 (Table S4), 8 out of 11 have been demonstrated as integral to MS pathogenic mechanisms (Fig. 3H). Together, these results demonstrate that M2 provides therapeutic options for MS.

3.3. Drug repurposing through the PRS-based approach

For the identified therapeutic module M2, drug repurposing based on PRS was subsequently performed (Fig. 4A), including weighted DTN construction, GNM modeling, and perturbation score calculation for each drug in the network. Based on the M2 module, the corresponding DTN was constructed by integrating 945 drugs, interconnected via 2,048 associations with the target proteins, accommodating 1,203 nodes and 6,328 edges (Fig. 4B). Then, DeepPurpose (Fig. S3A), a deep learning method, was used to predict pIC_{50} values for weighting each DTI within the DTN. These values, ranging from 1.02 to 43.39, are presented in Fig. S3B, with higher values indicating a stronger binding affinity.

In the GNM modeling of DTN, the PRS matrix was derived (Fig. 4C), in which the rows correspond to the effectiveness of nodes within the network, while the columns represent the sensitivity of nodes. Nodes with high sensitivity scores potentially have a greater impact on the overall network when perturbed, thereby playing a significant role in the network's dynamics and DTIs. Following the establishment of the PRS matrix, we proceeded with the computation of two perturbation scores: *ps* for representing the perturbation ability of each drug with its targeting proteins, while *PS*, the cumulative score of *ps*, for representing the perturbation ability of each drug in the context of the target module. The *ps* scores of the top 50 drugs with the highest perturbation abilities for the M2 are shown in Fig. 5. Consequently, the drug Fostamatinib, interacting with several target locations in the network including MAPK15, STK33, MAPK9, and others, emerges as a significant contributor. It seizes the top spot in our ranking with a notable score (*PS*) of 1.6909610, attesting to potentially impactful interactions within the network.

To provide a comparison with shortest-path-based methods, we initially applied a shortest-distance approach SAveRUNNER [35] to identify potential drug candidates, resulting in a selection of eight significant drugs (Table S4). However, this approach only yielded drugs without corresponding target proteins. To address this

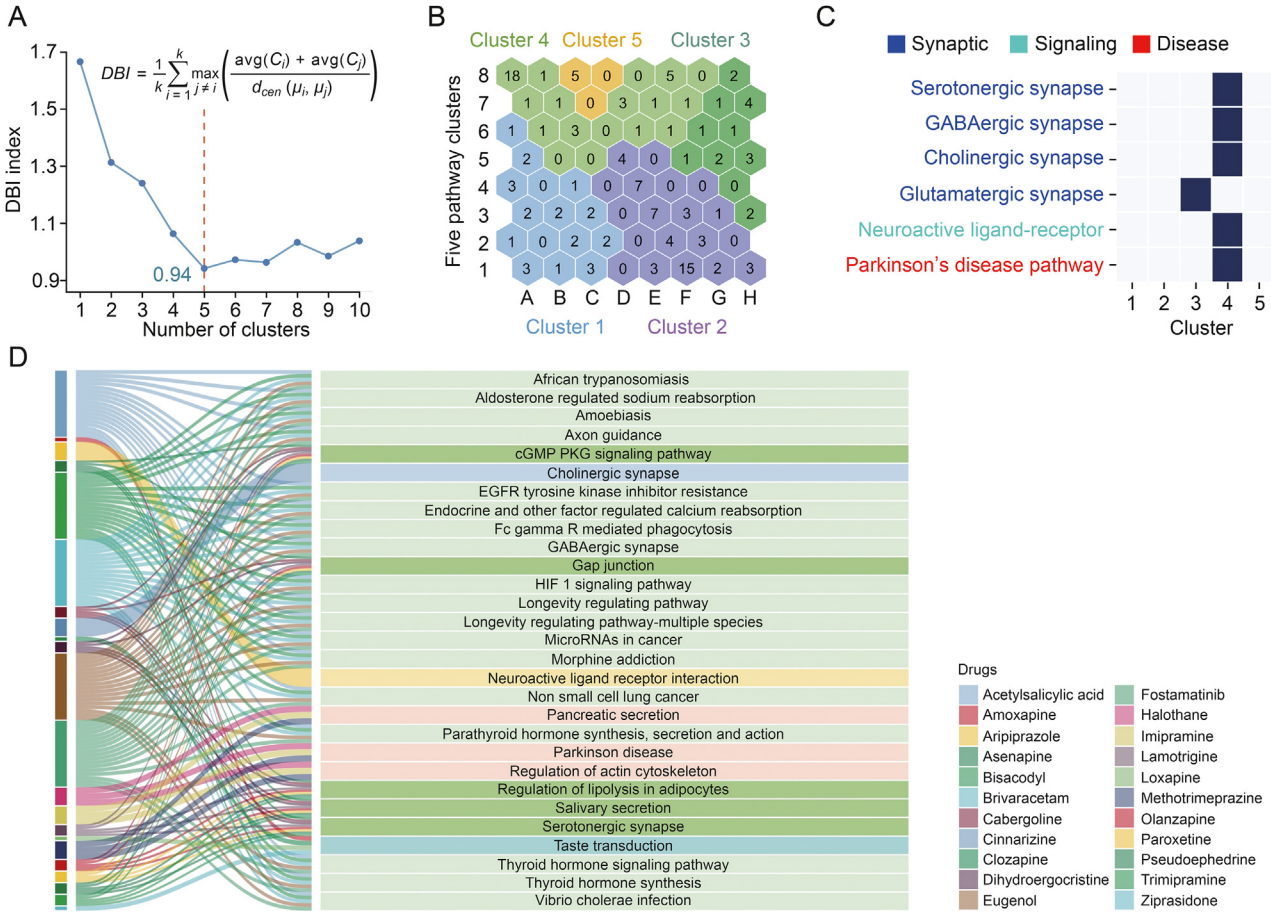


Fig. 6. Mechanism of action (MoA) analysis. (A) The line graph depicts the Davies-Bouldin index (DBI) values obtained using different numbers of clusters. This visualization reveals an optimal clustering delineation at five clusters, where the DBI value is the lowest at 0.94. (B) The self-organizing map (SOM) component plot displays five clusters selected using the U-matrix and DBI. The clusters of each SOM neuron can be distinguished by their respective colors. The number of pathways mapped to each neuron is represented by the displayed value within the neuron. (C) The heatmap of the interaction patterns across various MS-related pathways ('Synaptic', 'Signaling', and 'Disease') associated with different clusters. See Table S6 for the detailed information of pathways. (D) The relationship between drugs in Cluster 4 and their associated pathways. The drugs mapped to the corresponding neurons in Cluster 4 are shown on the left, and the pathways mapped to the corresponding neurons in Cluster 4 are shown on the right. The connection lines between the drugs and pathways depict the drug-pathway interactions within Cluster 4.

limitation, we extracted 21 DTIs involving these drugs within our network (Fig. S4A). We then performed molecular docking for these 21 pairs, alongside the top 21 pairs ranked by *PS* from our method. The docking results showed no significant differences between the two sets (Fig. S4B), indicating that the interactions identified by our method have comparable potential to those derived from the shortest-path-based approach. In addition, the blood-brain barrier presents a major challenge in treating CNS diseases, as it prevents most CNS drug candidates from crossing. Our study has showed that repurposed drugs selected based on our method generally have higher BBBP than those from the shortest path method (Tables 1 and S4), indicating that drugs identified by our approach are more likely to penetrate the brain and exert therapeutic effects. These findings suggest that our approach is capable of identifying high-potential drug-target interactions effectively.

3.4. Mechanism of action analysis

To further elucidate the mechanism of action (MoA) of the above 50 drugs (Fig. S5), the KEGG pathway enrichment analysis was performed (See Supplementary Methods), which reveals that the corresponding protein targets are significantly involved in 136 critical biological pathways. Next, we calculated precision, recall, and the F1 score to create a scoring matrix (Table S5), and used self-organizing map (SOM) analysis to reveal relationships between the drugs and pathways. We then visualized the data's density distribution with a U-matrix based on Euclidean distances, and proceeded with a cluster analysis on this U-matrix using the K-means algorithm. Byleveraging the Davies-Bouldin index (DBI) for cluster assessment, we determined that a total of five clusters, with a minimal DBI value of 0.94, optimally segmented the 136 pathways (Fig. 6A).

These pathways were classified into five clusters, with the 50 drugs mapped to each neuron (Fig. 6B and Table S6). As listed in Tables S5 and S6, drugs were mapped to 4 pathways in Cluster 1; 1 drug was mapped to 3 pathways in Cluster 2; 11 drugs were mapped to 9 pathways in Cluster 3; 22 drugs were mapped to 29 pathways in Cluster 4; 6 drugs were mapped to 5 pathways in Cluster 5. Among these drugs-associated pathways, we found that serotonergic synapse, GABAergic synapse, and cholinergic synapse were enriched. Alteration of synaptic plasticity and transmission have been recognized as critical determinants of early neurodegeneration in MS, and serotonergic neurotransmission is altered in limbic and paralimbic regions as well as in the frontal cortex in MS patients. We then performed a quantitative comparison of MS-related pathway associations across clusters, as shown in Fig. 6C. Cluster 4 was prioritized due to its prominent representation in the drug-pathway network, containing 22 drugs mapped to 29 pathways—the highest among all clusters. The enrichment of synaptic pathways (serotonergic, GABAergic, and cholinergic) in this cluster

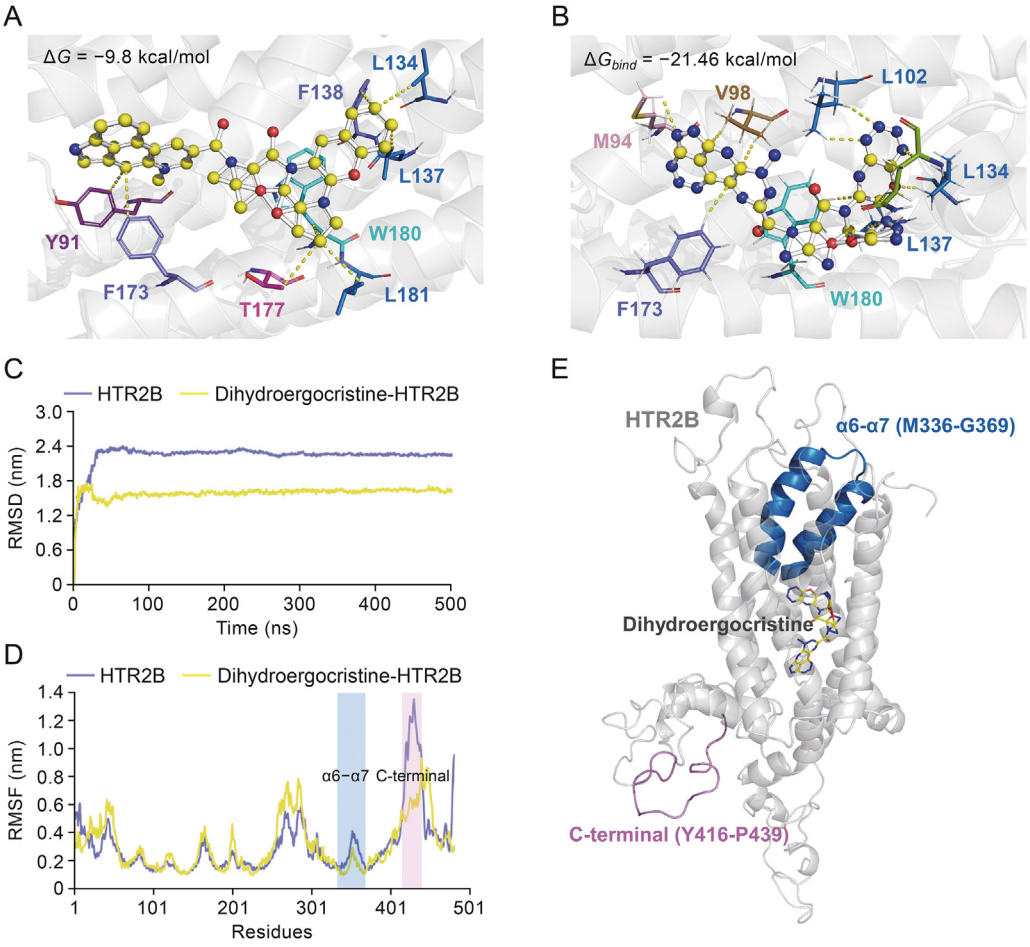


Fig. 7. Molecular docking and simulation of the dihydroergocristine-serotonin 2B receptor (HTR2B) complex. (A) The binding pose and affinity of the dihydroergocristine-HTR2B complex based on molecular docking. (B) The binding pose of the equilibrium state after 500 ns molecular dynamics (MD) simulation, and the binding free energy from MD trajectory. (C) Root mean square deviations (RMSDs) for apo and dihydroergocristine binding HTR2B during the MD simulations. (D) Root mean square fluctuation (RMSF) results for α-carbon atoms of apo and dihydroergocristine binding HTR2B systems, highlighting two regions become more stable after the dihydroergocristine binding (E).

aligns with the critical role of synaptic dysfunction in early MS neurodegeneration, offering potential therapeutic solutions tailored to address the complex nature of MS. The intricate relationships between the drugs in Cluster 4 and their associated pathways are depicted in Fig. 6D. Specifically, 22 drugs correspond to Cluster 4, and 13 of them have been explored in the literature for their potential as viable treatments for MS.

For the selection of novel drugs, a multi-step screening strategy was implemented based on the following criteria: 1) possession of a top 50 *PS* value; 2) original therapeutic indications for CNS diseases with specific targeting of Cluster 4; 3) absence of documented evidence for MS treatment. Accordingly, dihydroergocristine, a compound originally indicated for Alzheimer's disease (AD), emerged as a promising repurposing candidate for MS, which is indicated by targeting serotonin 2B receptor (HTR2B) to form the interaction with the highest *ps* (Fig. S6). For the possible structure-based MoA (See Supplementary Methods), dihydroergocristine binds with HTR2B using hydrophobic contacts of Y91, L134, L137, F138, F173, T177, W180, and L181, complemented by a π-stacking interaction with W180 (Fig. 7A), with the binding affinity of −9.8 kcal/mol. The all-atom 500 ns molecular dynamics (MD) simulation shows that the binding free energy of dihydroergocristine-HTR2B complexes became −21.46 kcal/mol (Fig. 7B). In addition, the stability of dihydroergocristine-HTR2B complexes was also studied using root mean square deviation (RMSD) and root mean square fluctuation (RMSF). As shown in Fig. 7C, the stable HTR2B system binding with dihydroergocristine shows much smaller RMSD (1.5 nm), compare with apo state with RMSD of 2.3 nm. Due to the RMSF comparison, as shown in Figs. 7D and E, the C-terminal region including Y416-P439, and the part of α6-α7 region including M336-G369 become more stable after the dihydroergocristine binding. This indicated that dihydroergocristine may be a repurposable drug targeting HTR2B in MS.

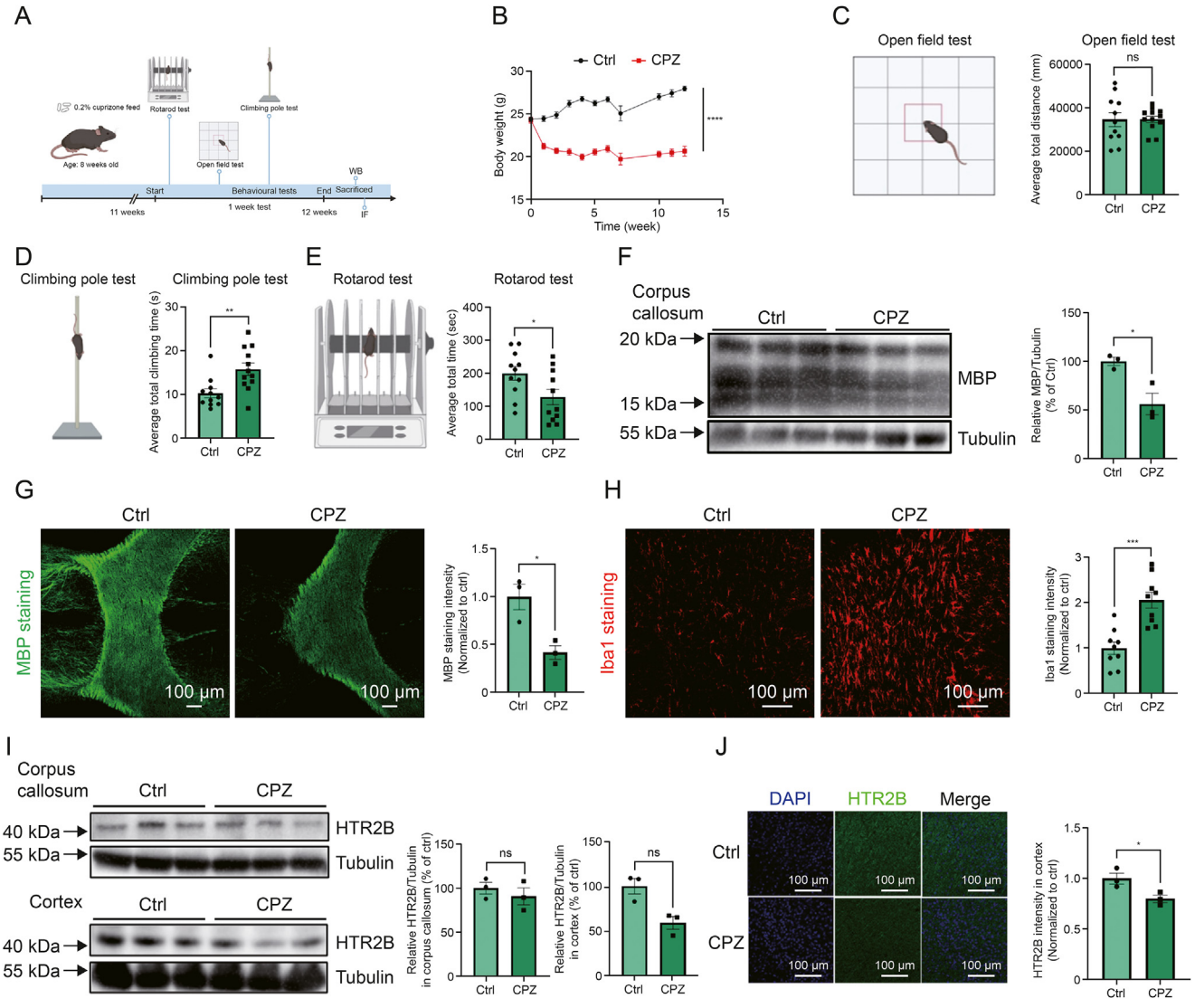


Fig. 8. Alteration of serotonin 2B receptor (HTR2B) in a cuprizone-induced chronic demyelination model. (A) Scheme profile of the cuprizone diet-induced chronic-myelinating mouse model. (B) Quantitative analysis of the weight of mice with a control diet or cuprizone diet each week. (C) Quantitative analysis of total distances of mice in the open field test. (D) Quantitative analysis of time to descend of mice in the pole test. (E) Quantitative analysis of latency to fall of mice in the rotarod test. (F) Representative Western blots (WB) and quantitative analysis of myelin basic protein (MBP) and tubulin proteins in the corpus callosum. (G) Representative images and quantitative analysis of MBP staining in the corpus callosum. (H) Representative images and quantitative analysis of ionized calcium-binding adapter molecule 1 (Iba1) staining of the corpus callosum. (I) Representative WB and quantitative analysis of HTR2B and Tubulin proteins in the corpus callosum and cortex. (J) Representative immunostaining and quantitative analysis of HTR2B protein in the cortex. $n = 11$ control chow (Ctrl) mice and 12 cuprizone chow (CPZ) mice in the behavioral studies (B–E), $n = 3$ mice/group (F–J). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$, ns: not significant. DAPI: 4′,6-diamidino-2-phenylindole.

3.5. Validation the alteration of HTR2B in the animal model of MS

To further investigate the role of HTRT2 in the MS, we firstly applied a 12-week-cuprizone-induced chronic demyelinating mouse model (Fig. 8A). The weight loss of cuprizone chow (CPZ) mice was observed a significant difference after one week (Fig. 8B). Moreover, CPZ has been known to cause the impairment of mouse motor coordination and balance. Therefore, a batch of behavior tests including open field test, rotarod test, and pole test were determined at the end of 12 week. In the open field test, no difference was observed for the total distance of mouse movement (Fig. 8C). A longer descending time on the pole and a shorter duration time on the rod were found for the CPZ mice compared to the control chow (Ctrl) mice in the pole test and rotarod test, respectively (Figs. 8D and E). These findings suggest that the CPZ mice may have reduced motor coordination and balance, as evidenced by their impaired performance on the pole and rotarod tests. After these behavior test, mice were immediately sacrificed and performed the immunoblotting and staining. Myelin basic protein (MBP) is considered as a pathological factor in the pathogenesis of MS. We first determined MBP protein level between Ctrl and CPZ mice by Western blotting. The results showed a dramatic reduction of MBP proteins in the CPZ mice compared to Ctrl mice (Fig. 8F). Moreover, MBP staining also demonstrated a prominent loss of MBP positive fibers in the corpus callosum of all CPZ-treated mice (Fig. 8G). To further confirm the CPZ-induced demyelination, we also evaluated other features induced by CPZ treatment. Extensive evidence shows that multiple sclerosis is a chronic immune-mediated neuro-inflammatory disease of the CNS. Microglia are the resident macrophages which play crucial roles in the progression of MS. We performed immunostaining for ionized calcium-binding adapter molecule 1 (Iba1), a cellular marker of microglia activation. The Iba1 staining showed that CPZ-treatment significantly increased the number of Iba1-positive cells in the corpus callosum (Fig. 8H). Together, it indicated the involvement of microglia activation in the CPZ-induced chronic demyelination.

To evaluate the potential alteration of HTR2B in the CPZ-induced demyelination, a Western blotting assay was performed to detect the HTR2B protein level in the corpus callosum and cortex of the mice. Our findings revealed a distinct reduction of HTR2B in the cortex, but not corpus callosum of mice that were treated with CPZ in comparison to the control (Fig. 8I). Moreover, the immunostaining in the cortex also demonstrated the significant decline of HTR2B protein in the CPZ-treated mice (Fig. 8J). These results suggest that HTR2B alteration is highly associated with MS progression.

4. Discussion and conclusion

Leveraging PRS, we introduced DTNPerturbome as a deep learning and network-based method for drug repositioning in the context of MS. DTNPerturbome employs shared genes between MS and other diseases as seed nodes to generate the MS comorbidity network through the RWR algorithm. The MS comorbidity network uncovers wider disease-relevant biological processes and signaling pathways related to MS, providing a more holistic view of the pathogenesis. Topological analysis and functional annotation of the MS comorbidity network have revealed that the “neurotransmission module”, associating with key neurological pathways as hallmarks of MS, was identified as the “therapeutic module” of MS. Then, the PRS strategy was further applied to study the drug response based on DTNs. This strategy allowed us to assess the potential impact of these top-ranked drugs on the “Neurotransmission Module” within the network.

MoA analysis suggested that dihydroergocristine is a repurposable drug of MS. Previous network-based drug repurposing strategies have identified various candidates for MS treatments. For instance, SAveRUNNER discovered that certain histaminergic drugs could serve as new potential therapeutic molecules for MS [42]. deepDTnet highlighted topotecan as a repurposable drug for MS by targeting retinoic acid receptor-related orphan receptor gamma t (ROR-γt) [43]. Most recently, using a gene regulatory network approach, the tyrosine kinase inhibitor sorafenib, the elastase (ELANE) inhibitor sivelestat, the histone deacetylase 1 (HDAC1) inhibitor vorinostat and the signal transducer and activator of transcription 3 (STAT3) inhibitor acitretin were predicted as four potential repurposable drug candidates [44]. Dihydroergocristine, a previously known inhibitor of γ-secretase [45], is demonstrated to be a serotonin (5-HT) receptor antagonist in the modulation of 5-HT neurotransmission [46]. In this study, we first demonstrate that dihydroergocristine, a constituent of Food and Drug Administration (FDA) approved drug, holds promise as a repurposable drug for MS by targeting HTR2B.

The extensive studies have shown that serum and platelet 5-HT levels are reduced in MS patients. It is known 5-HT plays a crucial role in the regulation via 5-HT receptors expressed on macrophages, dendritic cells, and T cells, which are key players in the immunological components of MS [47]. *In vitro* experiments showed that 5-HT caused monocyte polarization towards M2 macrophages by effectively suppressing LPS-induced pro-inflammatory cytokines and regulating M1 and M2 polarization-related genes [48]. HTR2B, a serotonin receptor primarily expressed by anti-inflammatory M2 macrophages, plays various crucial roles within the CNS, including the regulation of dopamine uptake, pulmonary vasoconstriction, and the modulation of cardiac structural function [49]. According to the human protein atlas database (https://www.proteinatlas.org/), it is evident that HTR2B is predominantly expressed in microglia. The unique expression pattern of HTR2B suggest that the different changes of HTR2B in the CPZ-induced demyelinating mouse model may signify the significant role of HTR2B in the pathogenesis of MS. Considering that the pathogenesis of MS involves abnormal immune system activation, dihydroergocristine may alleviate inflammatory responses and reduce the immune system's assault on the CNS, thereby slowing the progression of MS.

Throughout this study, we demonstrate the key contributions of DTNPerturbome as follows: 1) Regarding to network-based drug repurposing research, our method recognizes the importance of perturbation dynamics. Notably, DTNPerturbome extends the PRS analysis to the DTN level, which allows for the assessment of the network's dynamic response to external forces, offering new perspectives for predicting the role of drugs in networks, to facilitate the prioritization of candidate drugs. 2) Inspired by the disease comorbidity, our method begins by integrating shared genes between the target disease and other related diseases. By leveraging existing knowledge from multiple biomedical databases, our approach offers a comprehensive, network-based framework for drug repurposing, particular suited for rare diseases such as MS.

Our study has a few limitations. Perturbation-based repurposing approaches typically rely on datasets like the Connectivity Map and the LINCS project to identify potential drugs [4]. However, such datasets are limited for MS, making the lack of MS-specific perturbation data the first limitation of our method. Secondly, our approach depends on the existing knowledge of MS-associated genes and established PPIs data. The accuracy and comprehensiveness of these datasets directly affect our findings. Evaluating the robustness of our method across different PPI datasets, such as HIPPIE [50] or other high-quality curated databases [2], could

improve the reproducibility and reliability of our predictions. Lastly, when applying potential candidate repurposed drugs to the context of MS treatment, their dose prediction, safety, and efficacy still need to be further investigated through future experimental studies. In particular, the BBBP, as a critical determinant in CNS disease therapeutics, warrants thorough consideration in drug repurposing strategies. While dihydroergocristine demonstrates relatively low predicted BBBP, its established efficacy in AD treatment presents an intriguing paradox that necessitates further experimental validation. It is important to emphasize that although our current analysis specifically highlights dihydroergocristine, our methodological framework possesses the capability to systematically rank multiple drug candidates, thereby establishing a comprehensive predictive platform for drug repurposing applications. These identified limitations underscore the need for continued research efforts to enhance our understanding and overcome the therapeutic challenges associated with MS treatment.

In summary, by integrating a biophysics-based approach with network medicine methodologies, DTNPerturbome identified several potential drug candidates for the treatment of MS. Subsequent analyses, including pathway and structure-based MoA studies and validation in animal models, highlighted dihydroergocristine as a promising repurposable drug for MS, targeting HTR2B. This study not only offers a potential treatment for MS but also presents a methodology that can address translational medicine challenges in other complex diseases through drug repurposing.

CRediT authorship contribution statement

Yitan Lu: Formal analysis, Data curation. **Ziyun Zhou:** Writing – original draft, Formal analysis. **Qi Li:** Formal analysis. **Bin Yang:** Writing – review & editing, Validation, Investigation. **Xing Xu:** Investigation. **Yu Zhu:** Methodology. **Mengjun Xie:** Investigation. **Yuwan Qi:** Investigation. **Fei Xiao:** Validation. **Wenying Yan:** Validation. **Zhongjie Liang:** Writing – review & editing, Project administration. **Qifei Cong:** Writing – review & editing, Project administration, Funding acquisition. **Guang Hu:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Data and code availability

MS-related genes were downloaded from five public database: DisGeNET (https://www.disgenet.org/), DISEASES (https://diseases.jensenlab.org/), OMIM (https://www.omim.org/), PheGenI (https://www.ncbi.nlm.nih.gov/gap/phegeni/), and PGKB (https://www.pharmgkb.org/). The drug-target interaction data were downloaded from two public database, i.e., DrugBank (https://go.drugbank.com/), and Therapeutic Target Database (https://db.idrblab.net/ttd/). The shared genes of other 298 diseases were obtained from the Drug Repurposing Hub (https://clue.io/repurposing). The codes of DTNPerturbome are available at https://github.com/CSB-SUDA/DTNPerturbome.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpha.2025.101295.

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